

Whatsapp

Method 1 $\rightarrow$ Certified Authority
E2EE

MY CHOICE

Date: _____

Page No.: _____

Alice's Device

Bob's Device

1. Initial setup

Identity keys + Public keys

Authentication / Key verification

2. DH Key Agreement

Alice Pri-Key + Bob Pub-Key

$\downarrow$
DH / ECDH

$\downarrow$
Shared Secret

3. Root KDF

/HKDF

$\downarrow$

Initial Root Key

Alice sending $\rightarrow$

chain key

Bob receiving

$\downarrow$
chain KDF

$\downarrow$
chain KDF

$\downarrow$
Msg Key 1

$\downarrow$
Msg Key 1

AEAD / Encryption

$\downarrow$
Ciphertext 1

AEAD / Decryption

$\downarrow$
Msg 1

$\downarrow$
Discard Msg Key 1

Internet $\rightarrow$

Alice

Bob

4. Symmetric Ratchet

Chain Key
↓
KDF
↓
Msg Key

Chain Key
↓
KDF
↓
Msg Key

5. Asymmetric Ratchet Bob Replies

generate new DH Key

↓
DH / ECDH

↓
New shared secret

↓
Root KDF

Old Root Key
+
New DH secret

↓
New Root Key

↓
New chain

↓
Chain Key

↓
Msg Key

↓
AEAD encrypt

(Bob's new DH Public Key)
↓
Key + Ciphertext

Alice

← Internet

↓
extract Bob's new DH Pub Key

↓
DH / ECDH key agreement

↓
same new DH shared secret

↓
Root KDF → Old root + DH Secret

↓
New Root Key

↓
New receiving chain Key

Chain KDF



Msg key



AEAD Decryption



Bob's reply



Discard msg key